QISKIT FALL FEST 2026 · IIIT-DELHI · TRACK 3 QUANTUM SIMULATION · ADVANCED

Dynamics Under Noise

From an 8-spin Trotter study to a **certified 100-qubit kicked-Ising run** on IBM Heron QPU

100 qubits
error-aware line on heavy-hex,
0 SWAPs

1881 CZ
deepest circuit, 2q depth 38, 20 Floquet steps

3 exact checks
free fermions + Pfaffians · light cone · Clifford, all at N = 100

trust horizon
deepest depth where mitigated = exact

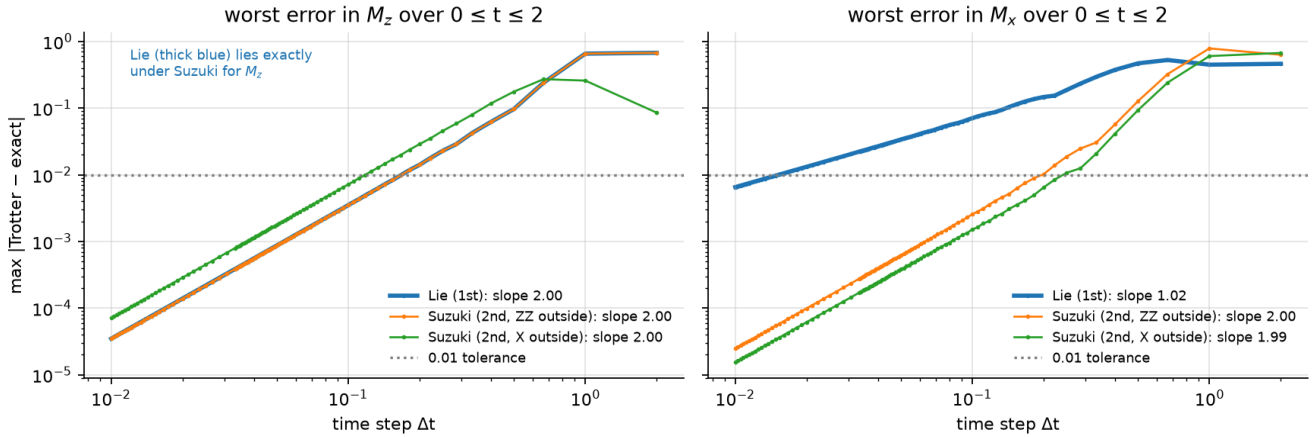
- **Beginner:** for Z observables from |0…0⟩, Lie-Trotter *is* Suzuki-Trotter, exactly. Mx needs 2nd order: 13× fewer CX for 0.01 accuracy.
- **Intermediate:** 2nd-order accuracy at 1st-order CZ cost. Our own TREX + ZNE on FakeFez noise. The noise-optimal Trotter step.
- **Advanced:** TREX + twirling + DD + ZNE on a real QPU, certified against exact answers at **every** depth without brute force.

[Team name] · [Member 1], [Member 2]

notebook: track3_dynamics_under_noise.ipynb

BEGINNER · 8 SPINS · J = h = 1 · t ≤ 2

Why the "right" Trotter step depends on what you measure



formula	obs	largest Δt (err<0.01)	CX Qiskit	CX merged

- $|0\ldots0\rangle$ is an eigenstate of every Z_iZ_{i+1}, so the **first ZZ layer is a phase**.
- A **final ZZ layer commutes** with every Z-basis observable.
- Strip both boundary pieces: Lie and Suzuki apply the **same gates** for M_z, $\langle Z_0Z_r\rangle$ (difference 1e-16). Lie is secretly 2nd order (slope 2.00).
- For M_x the final ZZ(Δt/2) is the correction: Lie is 1st order (slope 1.02), Δt = 0.015 vs 0.18.
- Merging adjacent half-steps: 2nd order at **1st-order CX cost**.

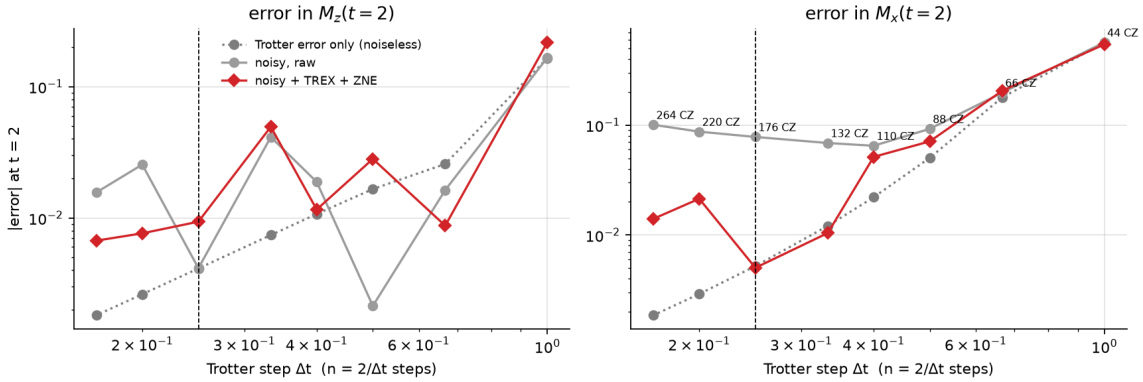
> Cheapest correct choice: Lie for Z-only; merged Suzuki (Δt = 2/13) for M_x.

INTERMEDIATE · CUT DEPTH BEFORE YOU MITIGATE

Hardware-aware compilation: every CZ removed is error you never mitigate

12-spin TFIM to t = 2 on FakeFez	CZ	2q depth	SWAPs	ESP

- **Brickwork ordering**: even bonds, then odd. 2q depth per step is constant in N.
- **Merged half-steps** and a **dropped trivial first layer**.
- **Error-aware line**: beam search over the coupling map minimising Σ CZ + readout + SX infidelity from live calibration. Finds a 100-qubit path on 156-qubit Heron.
- Kicked Ising at $\theta_J = -\pi/2$: each RZZ is **one CZ**. 100 qubits, 20 steps: 1881 CZ, 0 SWAPs.



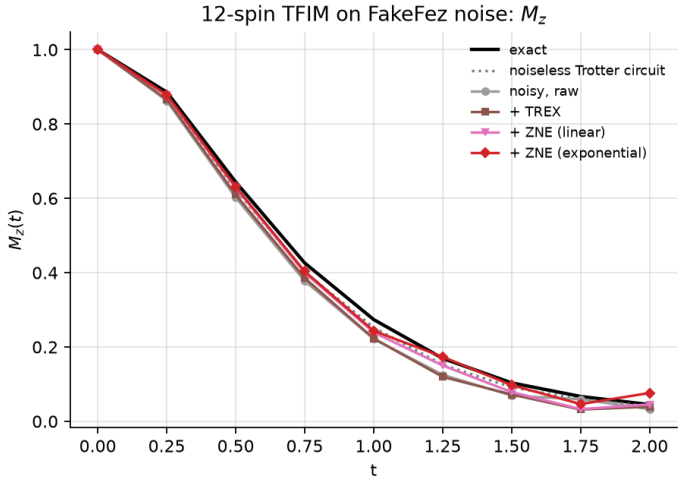
Noise-optimal step: total error = Trotter (∝Δt²) + noise (∝ #CZ ∝ 1/Δt). Δt = 0.25 beats the "Trotter-only" Δt ≈ 0.15 once noise is included.

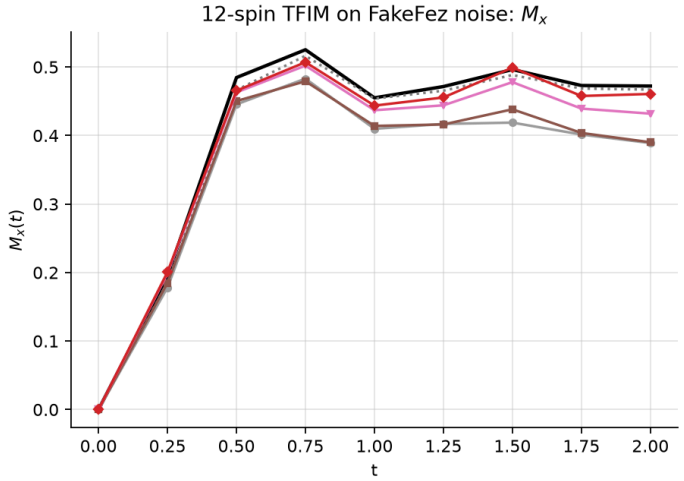
ESP = Π(1−ε_gate)·Π(1−ε_readout) from FakeFez calibration

§2.1, §2.4

INTERMEDIATE · 12 SPINS · FakeFez NOISE · OUR OWN MITIGATION CODE

Recovering the physics that noise hides





[i3_hardware_tfim.png: pending]

- **TREX**: random X before readout, undone classically, then rescaled by a twirled |0…0⟩ calibration.
- **ZNE**: every CZ folded to 3 or 5 CZs. Linear / exponential / Richardson to λ → 0.
- **Twirling** is a no-op on Pauli noise (we check it). It matters on hardware, where errors are coherent.
- Compact noise model checked against the full FakeFez model.

mean |error| vs exact	M_z	M_x

same 8 circuits re-run on the QPU (Job B) for the noisy-sim vs hardware comparison

§2.2–2.5

ADVANCED · HOW DO WE KNOW THE ANSWER AT 100 QUBITS?

Three independent exact certificates. No brute force.

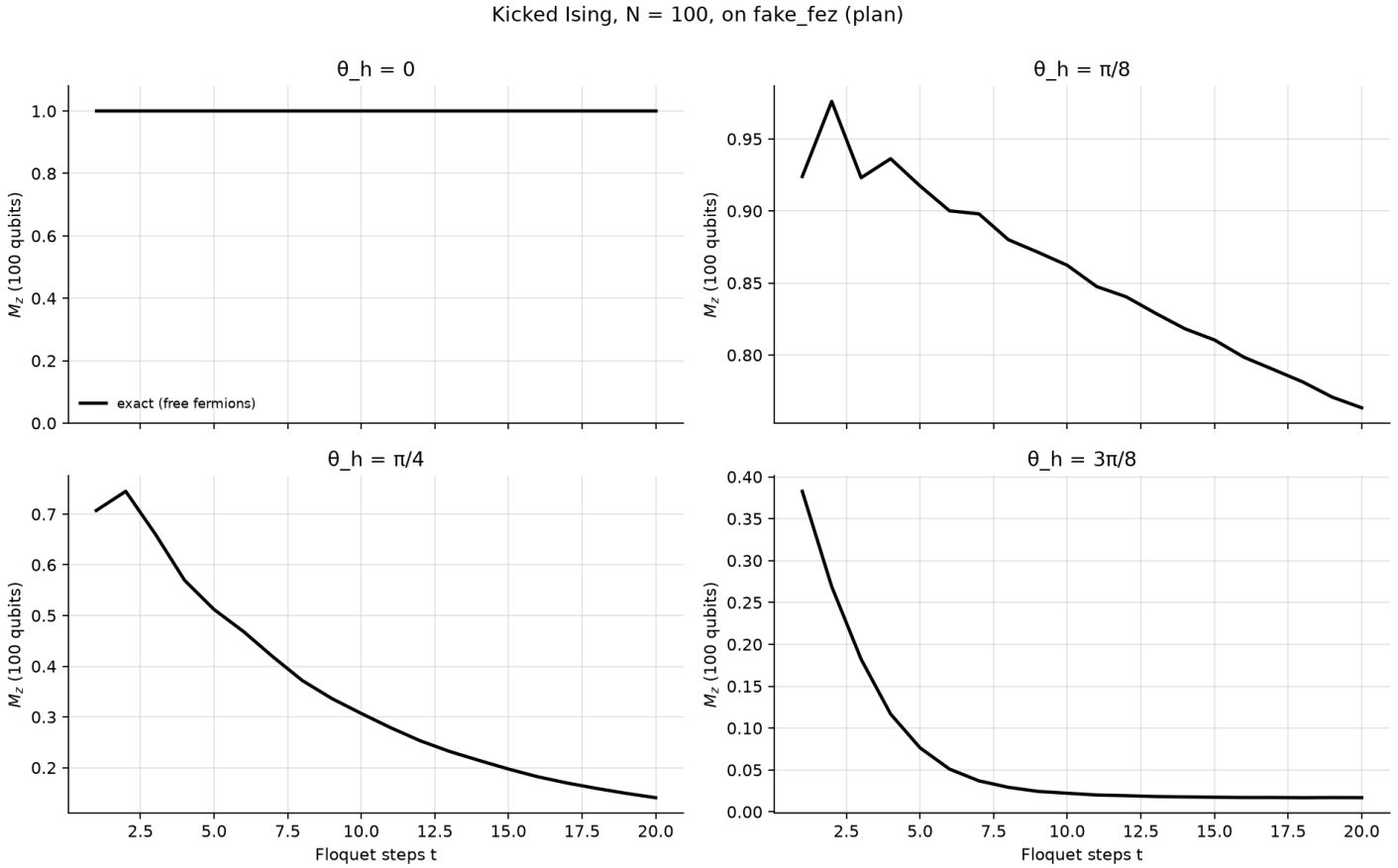
method	what it gives	cost	valid
Free fermions + Pfaffians JW in X basis, a ghost spin turns Z_i into an even Majorana string; Wick → Pfaffian	every $\langle Z_i \rangle$, any $\langle Z_i Z_j \rangle$	$O(N^3)$ · 0.3 s at N = 100, t = 20	integrable 1D kicked TFIM, **any depth**
Light-cone window $\langle Z_k(t) \rangle$ depends only on [k−t, k+t]; one (2t+2)-qubit edge window + reflection gives all 100 sites	every $\langle Z_i \rangle$, $\langle Z_i Z_{i+1} \rangle$	2^{2t+2}, independent of N	any local circuit, $t \lesssim 11$
Clifford point $\theta_h = \pi/2$ $S = U Z_c U^\dagger$, weight 2t−1 Pauli	$\langle S \rangle = +1$ exactly	stabiliser tableau	any N, any depth

- Cross-validated: free-fermion vs light-cone at N = 100 agree to **~1e-14**, and both match statevector at N = 12.
- Kicked Ising (Kim *et al.*, Nature 2023) on a 1D chain: $U_F = \Pi\, R_X(\theta_h) \cdot \Pi\, R_{ZZ}(-\pi/2)$.
- 52 circuits: $\theta_h \in \{0, \pi/8, \pi/4, 3\pi/8\} \times t \in \{1 \ldots 8, 10, 12, 14, 16, 20\}$, plus 7 Clifford circuits.

> Honest framing: integrability is *why* we can certify 100 qubits at depth 20. We use it as a calibrated benchmark of the device + mitigation stack, not as a claim of quantum advantage.

ADVANCED · 100-QUBIT HARDWARE RESULTS · IBM Heron QPU

Mitigated 100-qubit magnetisation vs the exact solution



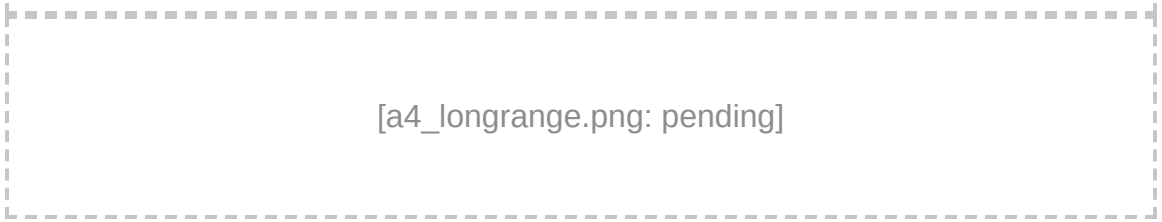
- **Raw**: SamplerV2, no mitigation.
- **λ = 1**: Pauli twirling + TREX + XpXm DD.
- **ZNE**: gate folding λ = 1, 3, 5, exponential extrapolation.
- Lines: exact free-fermion solution.

> Hardware results pending.

ADVANCED · CERTIFICATION & TRUST HORIZON

How far can we trust the device, and how would we know without an exact answer?







- **Clifford check**: a weight-(2t−1) stabiliser should read +1. Its decay measures the error per light-cone layer.
- **Trust horizon**: deepest t with |mitigated − exact| ≤ max(2σ, 0.05): pending.
- **Without an exact answer**: exponential vs linear ZNE agreement, Clifford-point decay, and light-cone windows for t ≤ 11.

raw bitstrings give $\langle Z_i Z_j \rangle$ at any distance after per-qubit readout inversion

§3.4–3.6

LIMITATIONS · NEXT STEPS · REPRODUCIBILITY

Where it works, where it breaks, what we would do next

- **Classically easy by design**: integrability lets us certify. A longitudinal field h_z breaks it, leaving only light cone + Clifford + consistency checks.
- **Gate-folding ZNE** multiplies depth by 3–5. At $t \geq 16$ (1500+ CZ) $\lambda = 5$ nears the noise floor and the fits disagree.
- **Pauli noise model** in simulation cannot show the value of twirling or DD (coherent errors), so we apply those on the QPU.
- **Next**: PEA-ZNE with learned layer noise (only 2 distinct CZ layers), fractional RZZ for the TFIM (halves depth), a non-integrable chain.

QPU job	primitive	QPU seconds
pending		

- Everything in one notebook that runs top to bottom (~5 min). QPU and long sweeps are cached in `results/`.
- `hardware/run_hardware.py`: dry-run / submit / fetch with the exact same circuits.
- Tutorial credit: BasQ Qiskit Fall Fest 2026 (B. Tirado). AI assistance: Claude, for coding and writing, with every number verified in-notebook.